CHM 43300 - Biochemistry

Instructor and Teaching Assistants

Instructor: Dr. Emmanouil Zacharioudakis, Assistant Professor, Chemical Biology
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Office Hours: 10-12 pm on Fridays at DRUG building, room 123

Teaching Assistants:
Stephanie Barrios - barrios6@purdue.edu (Office hours by appointment)
Ala Ahmad - ahmad119@purdue.edu (Office hours by appointment)

Lecture Time and Place

Lectures will be held at Brown Laboratory of Chemistry, room 1154, on Mondays, Wednesdays, and Fridays from 8:30 am-9:20 am.

Course Description

This course provides a comprehensive overview of biochemical principles, focusing on the structure and function of biological macromolecules, enzymatic mechanisms, and the regulation of metabolic pathways. It emphasizes integration of biochemical knowledge with applications in health, disease, and biotechnology. This course should serve as excellent preparation for higher-level biochemistry courses as well as standardized pre-health professional exams.

Learning Outcomes

By the end of this course, students will be able to:

1. Describe the structures and functions of proteins, carbohydrates, lipids, and nucleic acids.
2. Explain enzymatic mechanisms, kinetics, and regulation.
3. Understand the basics of how cells do the chemistry that is essential for life.
4. Understand the fundamental metabolic pathways essential for life, and their regulation.
5. Critically analyze biochemical data and literature.

Learning Resources

- Textbook (recommended): Biochemistry (tenth edition) by Jeremy Berg, Gregory J. Gatto Jr., Justin K. Hines, John L. Tymoczko, Lubert Stryer. Publisher: Macmillan.
- All course materials (lecture notes, assignments, grades) will be available via Brightspace.

Course Assessments

Assignments: 20%

Exam 1 - September 23, 2025 (8:00-9:30 PM): 20%, Location: WTHR 172

Exam 2 - October 28, 2025 (8:00-9:30 PM): 20%, Location: WTHR 172

Final Exam - December (assigned by Registrar): 40%

Course Assignments

There will be six course assignments during the semester. The assignments will be administered every two weeks on Friday. You must compose and write your own independent answers. Any papers with identical or nearly identical answers, or answers plagiarized from other sources (e.g. Artificial Intelligence resources) will be given automatically zero. The due date for every assignment will be 7 seven calendar days after it was administered. Late assignments will not be accepted.

Use of Artificial Intelligence Resources

Students are strongly advised to avoid using artificial intelligence (AI) resources when working on assignments. If students choose to use AI tools, these resources must not be used to generate solutions to problems or direct answers to assignment questions. Any response produced by AI and submitted as an answer will automatically receive a grade of zero**.** The only acceptable use of AI resources is as a supplementary aid to help students better understand the assignment instructions, questions, or problems—never to solve them.

Course Exams

There are three exams scheduled for this semester. See above the dates and time for the first two exams. The location for the third exam which is the final exam will be determined from the registrar office. All exams are in person. Any exams that have answers that are identical, nearly identical, or that have been plagiarized will be given automatically zero.

Grading Scale

A: ≥ 90%

B: 80-89%

C: 70-79%

D: 60-69%

F: < 60%

Course Schedule

Week	Topics (Chapters)	Assessments
Week 1	Introduction and Organic Chemistry Review (Chapter 1)	-
Week 2	Proteins (Chapter 2)	Assignment 1
Week 3	Protein Biochemistry (Chapter 3)	-
Week 4	DNA, Central Dogma, and Gene Regulation (Chapters 4, 5, 32 and 33)	Assignment 2
Week 5	Hemoglobin and Enzymes (Chapters 7 and 8)	Exam 1 (Sept 23)
Week 6	Enzymes and Catalysis (Chapters 8 and 9)	-
Week 7	Regulatory Mechanisms (Chapter 10)	Assignment 3
Week 8	Carbohydrates (Chapter 11)	-
Week 9	Lipids and Membranes, (Chapter 12)	Exam 2 (Oct 28)
Week 10	Channels and Pumps, and Basics of Metabolism (Chapters 13 and 15)	-
Week 11	Glycolysis & Gluconeogenesis	Assignment 5

	(Chapter 16)	
Week 12	Citric Acid Cycle (Chapter 17)	-
Week 13	Oxidative Phosphorylation (Chapter 18)	Assignment 6
Week 14	Review & Wrap-up	-
Week 15	Practice/questions	-
Finals Week	Final Exam	-

Course Policies and University Regulations

Regrade Policy

The exam answers must be written in ink for a regrade. Requests for regrades on exams and quizzes will be accepted only within 1 week after they are returned to the class, accompanied by a letter indicating explicitly which problems are potentially in error and what you feel the correct answer should be.

Attendance Policy

This course follows Purdue's academic regulations regarding attendance, which states that students are expected to be present for every meeting of the classes in which they are enrolled. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, the student should inform the instructor of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to the instructor is not possible, the student should contact the instructor as soon as possible by email. When the student is unable to make direct contact with the instructor and is unable to leave word with the instructor's department because of circumstances beyond the student's control, and in cases falling under excused absence regulations, the student or the student's representative should contact or go to the Office of the Dean of Students website to complete appropriate forms for instructor notification.

Academic Guidance During Illness

If you're sick, STAY HOME. If you have a personal situation that may require you to miss class, please let the instructor know. We will work with you to ensure you are able to meet all the course objectives. As part of this effort, all course material and lectures will be available on Brightspace.

Classroom Guidance Regarding Protect Purdue

Any student who has substantial reason to believe that another person is threatening the safety of others by not complying with Protect Purdue protocols is encouraged to report the behavior to and discuss the next steps with their instructor. Students also have the option of reporting the behavior to the Office of the Student Rights and Responsibilities.

Academic Integrity

CHM43300 will adhere to University guidelines. Please read http://www.purdue.edu/odos/osrr/academicintegritybrochure.php for more information. Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778.

Nondiscrimination Statement

We are committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential.

Accessibility

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let the instructor know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Mental Health

• The Department of Chemistry has put together a useful site with common warning signs and where to go for help: https://www.chem.purdue.edu/health/

• If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try https://www.purdue.edu/caps/students/resources/self-help/digital/tao.php

• If you need support and information about options and resources, please contact or see the Office of the Dean of Students (765-494-1747).

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email.

Basic Needs Security

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is

no appointment needed and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday.

Modified Attendance, Deadline, and Testing Accommodations

Students with approved accommodations by the Disability Resource Center should contact the instructor via email and accommodations will be handled individually.

Grief Absence Policy for Students (GAPS)

If you experience the death of a family member or close friend, fill out the form at https://www.purdue.edu/advocacy/students/absences.html.

Military Absence Policy for Students (MAPS)

If you are required to complete mandatory military training, fill out the form at https://www.purdue.edu/advocacy/students/absences.html.

Medical Excused Absence Policy for Students (MEAPS)

Students may occasionally have to miss class and other academic obligations due to hospitalization, emergency department or urgent care visits, whether physical or mental health related in nature. The intention of this policy is to afford arrangements to students experiencing serious and short-term medical situations which cause them to miss coursework and/or exams.

Professional Development Activities

Absences due to professional development activities are evaluated on an individual basis and may be approved as excused. Documentation should be submitted to the instructor as soon as it is available, but no later than 1 week prior to the absence.

Required University Sponsored Activities (NCAA Athletics)

Students participating in NCAA Athletics or other required university sponsored activities must provide travel letters or other documentation to the instructor no later than 1 week prior to the absence.

Disclaimer

This syllabus is subject to change. You will be notified of any changes as far in advance as possible via an announcement on Brightspace. Monitor your Purdue email daily for updates.